

TPG4190 Seismic data acquisition and processing

Deconvolution

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Convolutional model

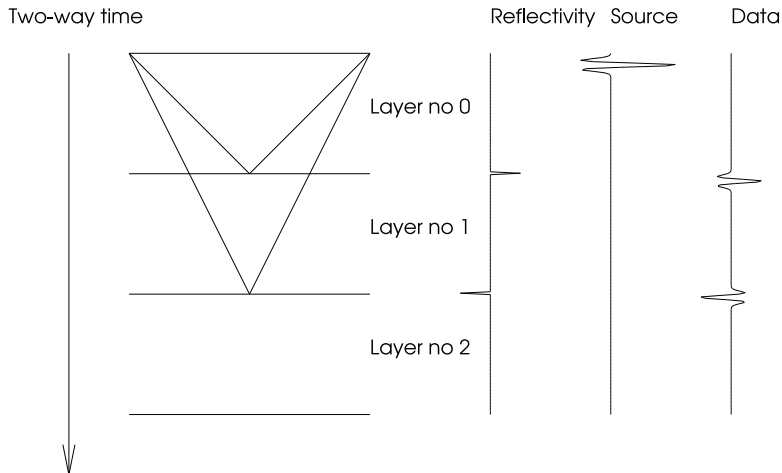


Figure: Layered earth model

Convolutional model

- ▶ vertical propagation

Reflected signal

$$a(t) = r_0 \frac{s(t - \phi_0)}{2d_0}. \quad (1)$$

- ▶ $\phi_0 = 2d_0/c_0$: two-way traveltime
- ▶ r_0 : reflection coefficient
- ▶ d_0 : layer thickness

Convolutional model

Reflection coefficient defined by

$$r_0 = \frac{c_1 \rho_1 - c_0 \rho_0}{c_1 \rho_1 + c_0 \rho_0}. \quad (2)$$

- ▶ c_0, c_1 : Wave velocity
- ▶ ρ_0, ρ_1 : Density

Convolutional model

- Signal recorded at the surface:

$$a(t) = r_0 \frac{s(t - \phi_0)}{2d_0} + r_1 \frac{s[t - (\phi_0 + \phi_1)]}{2d_0 + 2d_1}, \quad (3)$$

Convolutional model

- ▶ Neglect spherical spreading
- ▶ General model

$$a(t) = \sum_{k=0}^N r_k s(t - \tau_k), \quad (4)$$

- ▶ $\tau_k = \sum_{l=0}^k \phi_l$ two way traveltime to interface no k.

Convolutional model

► $\tau_k = k\Delta t.$

$$a_k = \sum_{l=0}^N r_l s_{k-l}, \quad (5)$$

$$\begin{aligned} a_k &= a(t = k\Delta t), \\ s_k &= s(t = k\Delta t). \end{aligned} \quad (6)$$

Continuous case:

$$a(t) = \int_{\tau=0}^{+\infty} r(\tau) s(t - \tau), \quad (7)$$

Convolutional model

- ▶ Vertical wave propagation
- ▶ Neglect vertical spreading
- ▶ Only primary reflections
- ▶ Earth is a linear filter
- ▶ $r(t)$ Earth's impulse response

Deconvolution

- ▶ Convolutional model for seismic waves

$$y(t) = r(t) * s(t), \quad (8)$$

- ▶ $s(t)$: seismic source
- ▶ $r(t)$: reflectivity

Deconvolution

- ▶ Recover $r(t)$ when
- ▶ $s(t)$ given
- ▶ $y(t)$ given

Spiking deconvolution

- Fourier transformation of (8)

$$Y(f) = R(f)S(f) \quad (9)$$

- Solution

$$R(f) = \frac{Y(f)}{S(f)}. \quad (10)$$

Deconvolution

- ▶ Equation (10) slightly reformulated

$$R(f) = S^{-1}(f)Y(f), \quad (11)$$

- ▶ $S^{-1} = 1/S(f)$: *inverse* filter.
- ▶ S^{-1} removes the source
- ▶ recovers reflection coefficients

Deconvolution

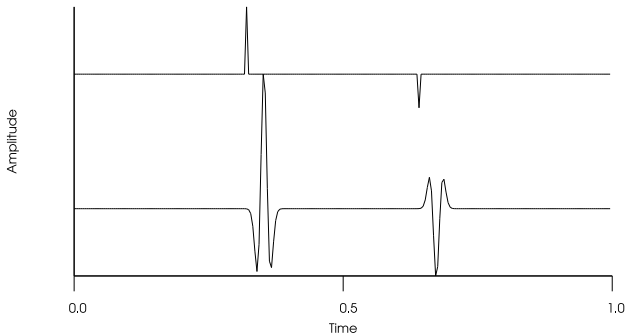


Figure: Spiking decon

The spectrum of a Ricker pulse and it's inverse

Deconvolution

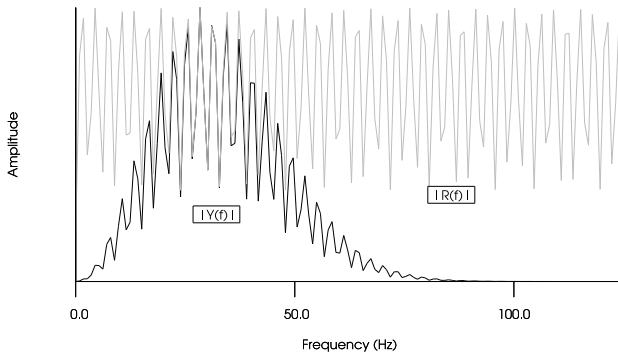


Figure: Deconvolution spectrum

The spectrum of the input data $|G(f)|$ and the deconvolved output $|R(f)|$.

Deconvolution

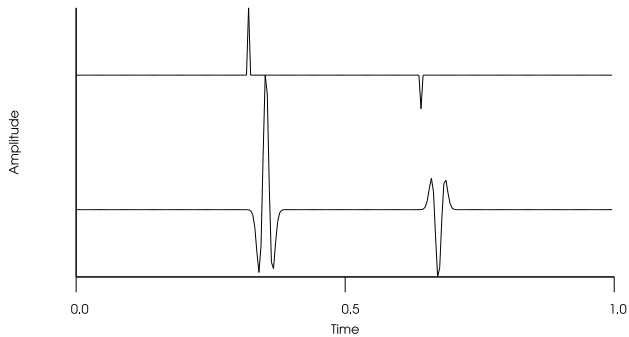


Figure: Deconvolution

Deconvolution

- Real seismic data contains noise

$$Y(f) = R(f)S(f) + N(f), \quad (12)$$

- $N(f)$: White (random) noise

$$R(f) = S^{-1}(f)Y(f) - S^{-1}(f)N(f). \quad (13)$$

Deconvolution

$S^{-1}(f)$ modified:

$$S^{-1}(f) = \frac{S^*(f)}{|S(f)|^2 + \epsilon}, \quad (14)$$

where ϵ is a small constant.

Inserting equation (14) into equation (13):

$$R(f) = \frac{S^*(f)}{|S(f)|^2 + \epsilon} Y(f) - \frac{S^*(f)}{|S(f)|^2 + \epsilon} N(f). \quad (15)$$

Increase ϵ to become much larger than $|s(f)|^2$

$$R(f) \approx \left(\frac{1}{\epsilon}\right) S^*(f)[Y(f) - N(f)]. \quad (16)$$

match-filtering is a stable approach for deconvolution.

Deconvolution

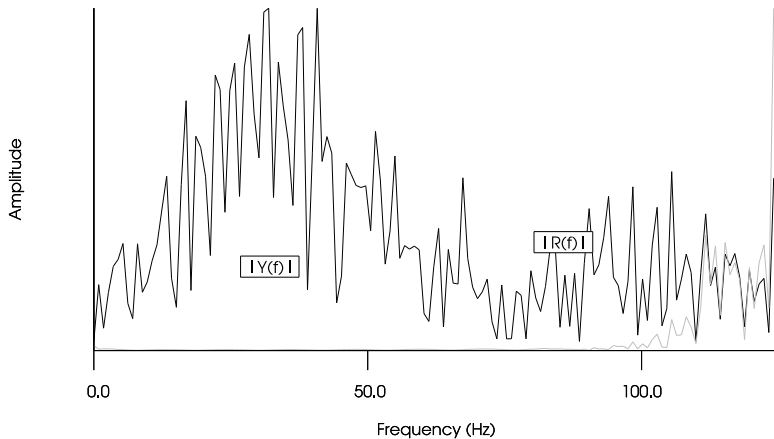


Figure: The spectrum of the input data with added noise $|Y(f) + N(f)|$ and the deconvolved output $|R(f)|$ (gray line).

Deconvolution

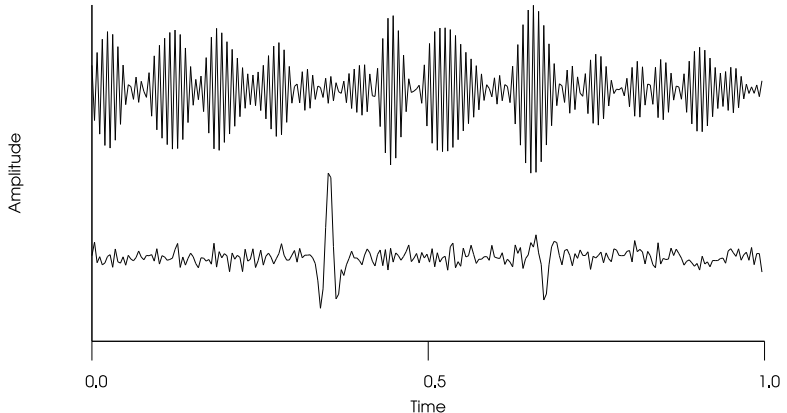


Figure: The input seismic data with added noise is shown in the lower trace, while the output of a spiking deconvolution filter applied to the input data is shown in the upper trace.

Deconvolution

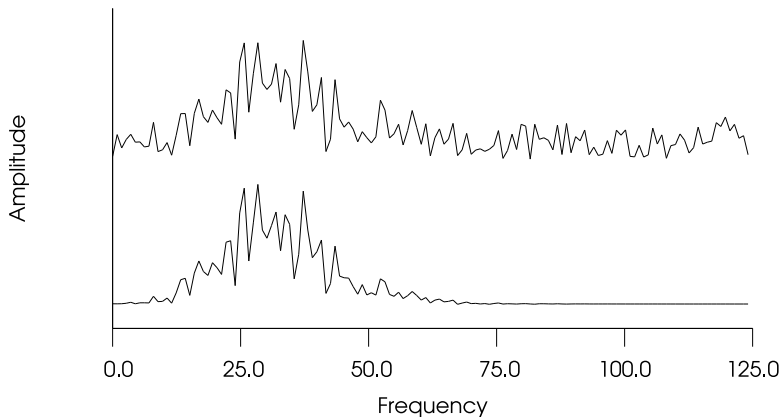


Figure: The spectrum of the input data with added noise $|Y(f) + N(f)|$ and the stabilized deconvolved output $|R(f)|$ (gray line).

Deconvolution

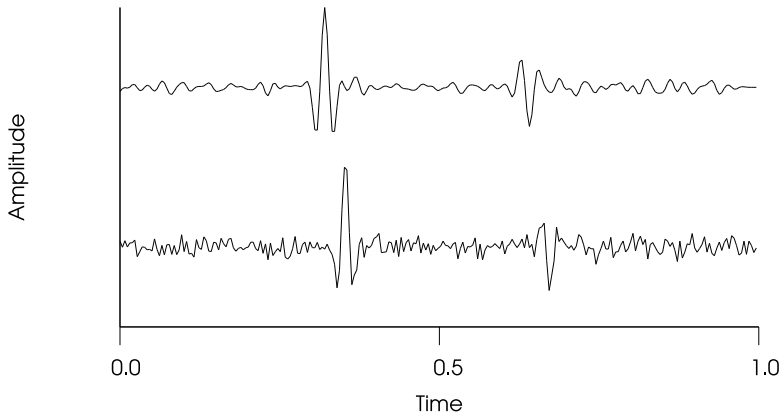


Figure: The input seismic data is shown in the lower trace, while the output of a match filter applied to the input data is shown in the upper trace.